Tri-Output Programmable POS & NEG Voltage AMOLED Bias

■ General Description

OCP21351 is highly integrated power solution for AMOLED, especially on wearable devices. By taking the input supply voltage, 2.9V to 4.6V, from battery, OCP21351 generates the AVDD, OVDD, and OVSS outputs following the target settings.

OCP21351 is operated in either symmetry or asymmetry mode, which is defined by SAS pin. And this SAS pin should be fixed before OCP21351 starts operating, before EN goes High level.

In Symmetry mode, while SAS is set as Low level, the initial output voltages of AVDD, OVDD and OVSS are 3.3V, 3.3V and –2.2V respectively. In Asymmetry mode, while SAS is set as High, the initial output voltages of AVDD, OVDD and OVSS are 2.8V, 4.6V and –2.2V respectively.

OCP21351 also integrates Serial Data Interface for the flexibility of the various settings. However, only OVSS and AVDD setting is allowed to be changed in Asymmetry.

Under voltage lock out, short circuit protection, and thermal shutdown are all built in OCP21351.

The OCP21351 is available in WLCSP-21B package to achieve optimized solution for PCB space.

■ Features

- Input Supply Voltage Range: 2.9V to 4.6V
- SRL Interface Protocol
- AVDD: 2.8V (SAS = high) to 3.3V (SAS = low)
- OVDD default: 4.6V (SAS = high) or 3.3V (SAS = 0)
- OVSS: Programmable −0.6V to −5.0V
- SRL Idle Mode for Light Load Operation
- Protection: UVLO, UVP, OTP
- WLCSP-21B Package, 2.8mmx1.2mm

■ Applications

- 2-4’AMOLED Bias in Portable Device

■ Pin Configuration

WLCSP-21B (Top View)

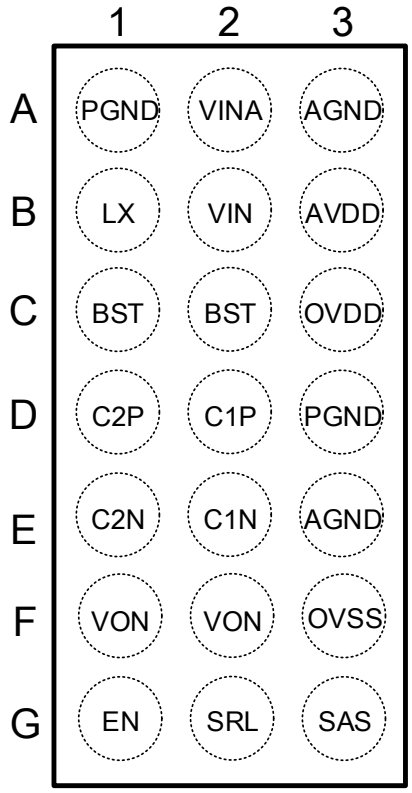
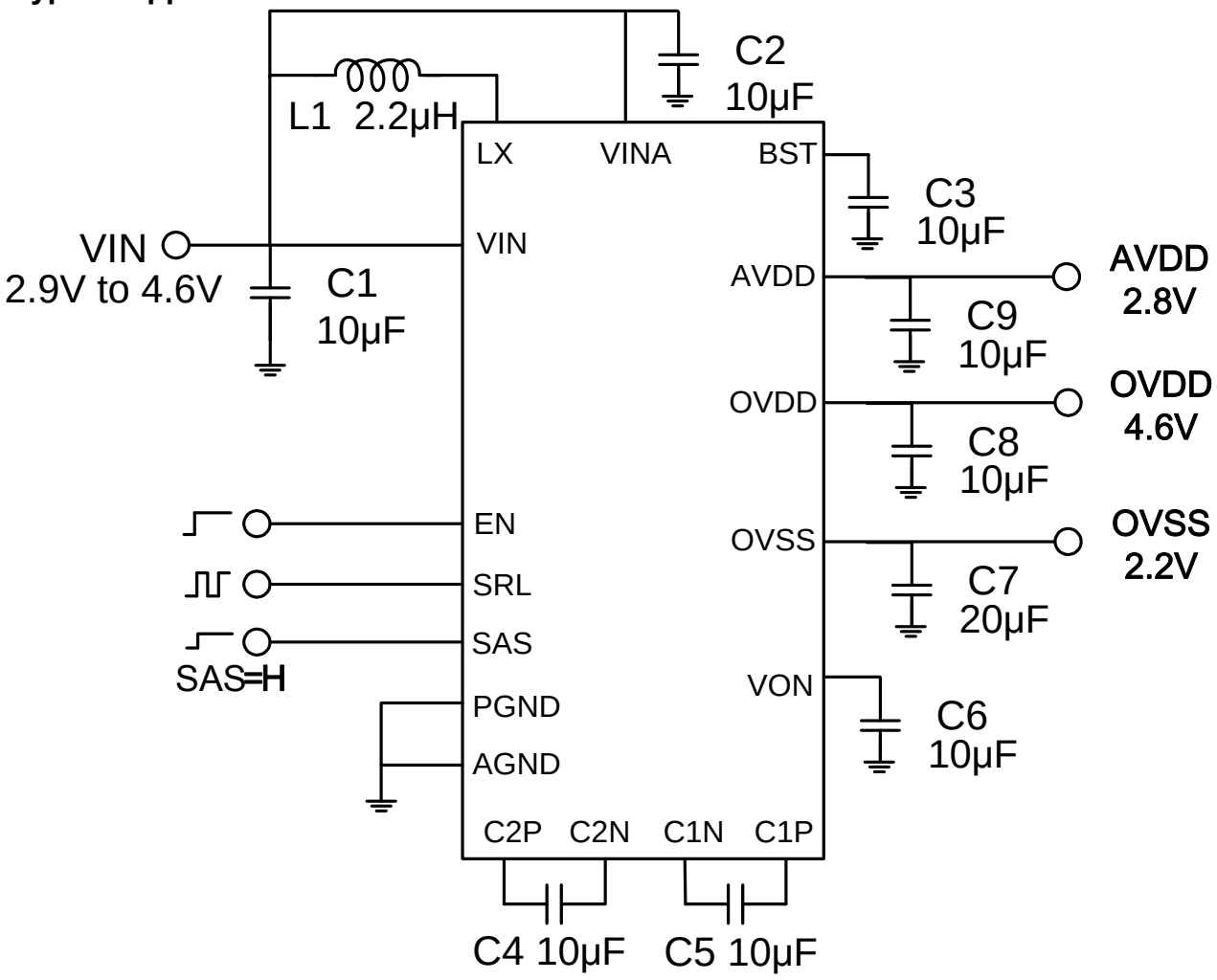


Figure 1, Pin Assignments of OCP21351

Pin Name	Pin No. WLCSP-21	I/O	Pin Function
A1	PGND	-	Power Ground
A2	VINA	I	Power Supply Input for LDO1
A3	AGND	-	Analog Ground
B1	LX	I/O	Switching Node of Boost Regulator
B2	VIN	I	Power Supply Input
B3	AVDD	O	Output Voltage for Driver
C1	BST	O	Output of Boost Regulator
C2	BST	O	Output of Boost Regulator
C3	OVDD	O	Positive Output Voltage for OLED Bias
D1	C2P	I	Flying Capacitor Connection
D2	C1P	I	Flying Capacitor Connection
D3	PGND	-	Power Ground
E1	C2N	I	Flying Capacitor Connection
E2	C1N	I	Flying Capacitor Connection
E3	AGND	-	Analog Ground
F1	VON	O	Negative Charge Pump Output
F2	VON	O	Negative Charge Pump Output
F3	OVSS	O	Negative Output Voltage for OLED Bias
G1	EN	I	Enable Pin
G2	SRL	I	One-Pin Serial Interface Control
G3	SAS	I	Symmetry or Asymmetry Power Selected Pin

■ **Typical Application Circuit**



Asymmetry

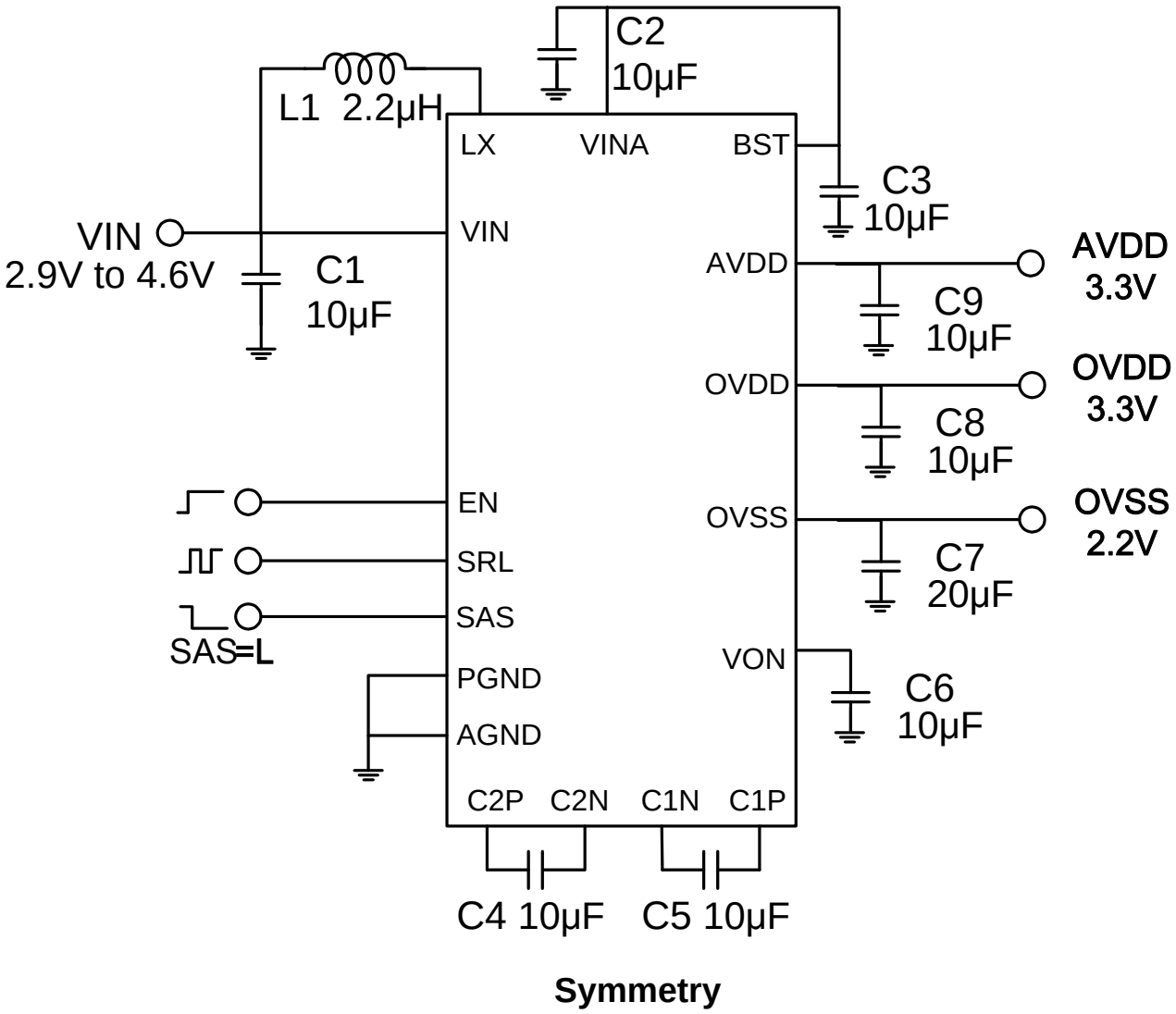


Figure 2, Typical Application Circuit of OCP21351

Block Diagram

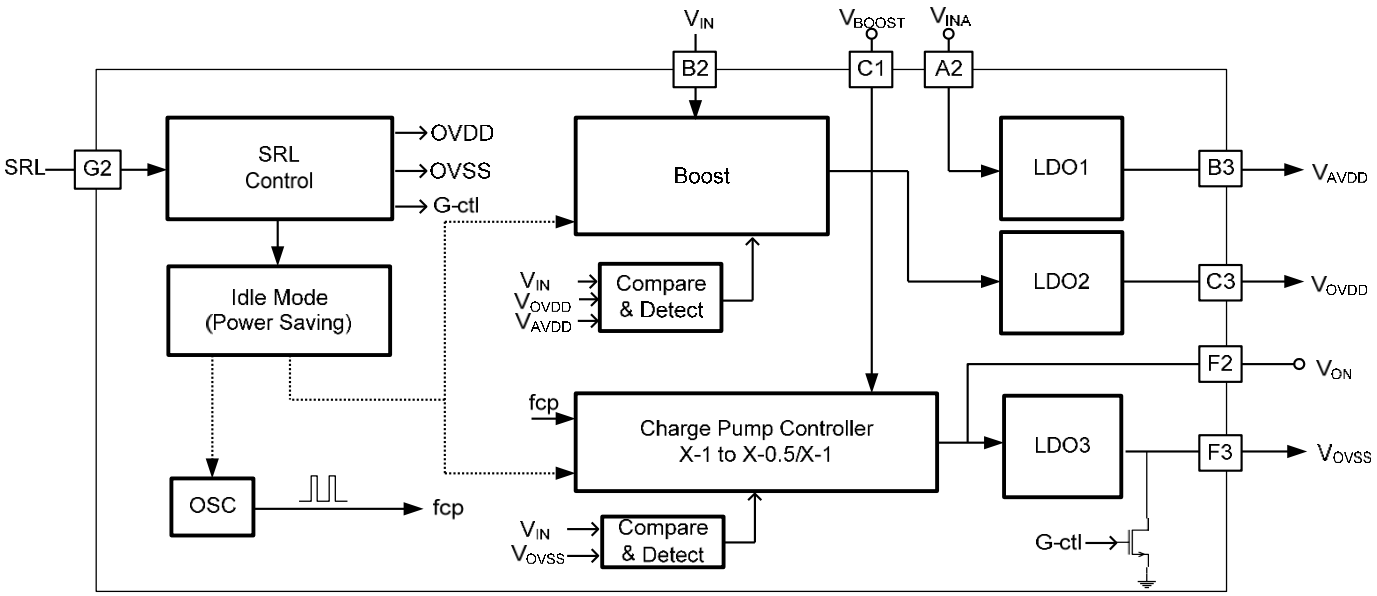


Figure 3, Block Diagram of OCP21351

■ Absolute Maximum Ratings[1] (TA=25℃, unless otherwise noted)

Parameter	Symbol	Rating	Unit
V_{IN} ， LX Pin to GND	LX,V_{IN}	-0.3 to +6.0	V
Output Voltage 1 (BST, AVDD, OVDD, C1P, C2P)	Vo1	-0.3 to +5.5	V
Output Voltage 2 (OVSS, VON, C1N, C2N)	Vo2	-5.5 to +0.3	V
All Other Pins to GND	-	-0.3 to V_{IN} + 0.3	V
Storage Temperature Range	T_S	-55 to +150	℃
Operating Junction Temperature Range	T_J	-40 to +150	℃
Maximum Soldering Temperature (at leads, 10 sec)	T_{LEAD}	300	℃

■ Recommended Operating Conditions[2]

Parameter		Symbol	Rating	Unit
V_{IN} Pin Voltage to GND		V_{IN}	+2.9 to +4.6	V
Operating Temperature Range		T_{OP}	-40 to +85	℃
Maximum Thermal Resistance	WCLSP-15	θ_{JA}	75	℃/W

Note: 1:Stresses above those listed in absolute maximum ratings may cause permanent damage to the device. Functional operation at conditions other than the operating conditions specified is not implied. Only one absolute maximum rating should be applied at any one time.

2: The device is not guaranteed to function outside of its operating conditions.

■ Electrical Characteristics

(V_{IN} = 2.9V to 4.6V, T_A = 25℃ , unless otherwise specified.,V_{IN} = 3.7V, SAS = 0, SRL=pulse7, V_{AVDD} = 3.3V, V_{OVDD} = 3.3V, V_{OVSS} = −3.3V, C_{INA} = C_{fly1} = C_{fly2} = C_{VIN} = C_{AVDD} = C_{OVDD} = C_{BST} = C_{VON} = 10μF, C_{OVSS} = 20μF)

Operating Power

PARAMETER	SYMBOL	TEST CONDITION	MIN	TYP	MAX	UNIT
V_{IN} Input Voltage Range	V_{IN}		2.9	3.7	4.6	V
Under Voltage Lockout Threshold	V_{UVLO}	Rising	2.5	2.6	2.7	V
		Falling	2.1	2.2	2.3	
Idle Mode Total Input Current	I_{IDLE_IN}	V_{IN} = 3.7V; SAS= 0 SRL= 0 & AVDD No Load	-	-	200	μA
Efficiency 3 SAS Low (I_{AVDD} = 3mA; I_{OVDD} = I_{OVSS} = 30mA)	Eff3	V_{IN} = 3.2V	-	81	-	%
		V_{IN} = 3.7V	-	84	-	%
		V_{IN} = 4.2V	-	77	-	%
Shutdown Current	I_{SD}	EN = SRL = 0	-	-	1	μA
EN, SAS High Voltage Level	V_{IH}		1.2	-	-	V
EN, SAS Low Voltage Level	V_{IL}		-	-	0.4	V
EN Pull Low Resistance	R_{EN}		-	0.5	-	MΩ
Thermal Shutdown	T_{SH}		-	150	-	℃
Thermal Shutdown Hysteresis	T_{Hys}		-	20	-	℃

■ Electrical Characteristics

(V_{IN} = 2.9V to 4.6V, T_A = 25 ℃, unless otherwise specified. V_{IN} = 3.7V, SAS = 0, SRL=pulse7, V_{AVDD} = 3.3V, V_{OVDD} = 3.3V, V_{OVSS} = −3.3V, C_{INA} = C_{fly1} = C_{fly2} = C_{VIN} = C_{AVDD} = C_{OVDD} = C_{BST} = C_{VON} = 10μF, C_{OVSS} = 20μF)

Synchronous Current Mode Boost Regulator

PARAMETER	SYMBOL	TEST CONDITION	MIN	TYP	MAX	UNIT
BOOST Output Voltage	V_{BOOST}		-	-	5.5	V
BOOST Maximum Load Current	I_{BOOST_MAX}	V_{OVDD} = 3.3V, V_{OVSS} = –3.3V	110	-	-	mA
		V_{OVDD} = 4.5V, V_{OVSS} = –4.5V	170	-	-	mA
LX Switching Frequency	f_{OSC_LX}		-	0.5	-	MHz
LX Current Limit	I_{LIM_LX}		0.4	0.6	0.8	A
LX NMOS ON-Resistance	R_{ON_NMOS1}		-	0.2	-	Ω
LX PMOS ON-Resistance	R_{ON_PMOS1}		-	0.2	-	Ω
LX Maximum Duty Cycle	D_{MAX_LX}		-	87	-	%
Soft Start Time	t_{SS1}		-	2	-	ms

■ Electrical Characteristics

(V_{IN} = 2.9V to 4.6V, T_A = 25℃ , unless otherwise specified. V_{IN} = 3.7V, SAS = 0, SRL=pulse7, V_{AVDD} = 3.3V, V_{OVDD} = 3.3V, V_{OVSS} = −3.3V, C_{INA} = C_{fly1} = C_{fly2} = C_{VIN} = C_{AVDD} = C_{OVDD} = C_{BST} = C_{VON} = 10μF, C_{OVSS} = 20μF)

AVDD

PARAMETER	SYMBOL	TEST CONDITION	MIN	TYP	MAX	UNIT
Output Voltage	V_{AVDD}	SAS = 0	2.6	3.3	3.6	V
Output Voltage		SAS = high	2.6	2.8	3.6	V
Output Voltage Tolerance		V_{AVDD}(default) I_{AVDD} = 10mA	–1	-	+1	%
Load Current Capacity	I_{AVDD}	SAS = 0 AVDD = 3.3V, OVDD = 3.3V, OVSS = –3.3V SAS = high AVDD = 2.8V, OVDD =4.6V, OVSS = –2.2V	10	-	-	mA
Line Regulation		V_{IN} = 2.9V to 4.5V I_{AVDD} = 10mA	-	5	10	mV
Load Regulation		V_{IN} = 3.7V I_{AVDD} = 0mA to 10mA	-	5	10	mV
Output Transient Ripple		V_{IN} = 3.7V SAS = 0 , AVDD=3.3V, OVDD= 3.3V, OVSS = –3.3V I_{AVDD} & I_{OVDD} & I_{OVSS} ≤ 5mA V_{IN} = 3.7V SAS = high , AVDD=2.8V, OVDD= 4.6V, OVSS = –2.2V I_{AVDD} & I_{OVDD} & I_{OVSS} ≤ 5mA	–10	-	+10	mV
Output Voltage Ripple		V_{IN} = 3.7V, SAS = 0 AVDD = 3.3V, I_{AVDD} ≤5mA OVDD = 3.3V, I_{OVDD} ≤ 30mA OVSS = –3.3V, I_{OVSS} ≤ 30mA V_{IN} = 3.7V, SAS = high AVDD = 2.8V, I_{AVDD} ≤ 5mA OVDD = 4.6V, I_{OVDD} ≤ 30mA OVSS = –2.2V, I_{OVSS} ≤ 30mA V_{IN} = 3.7V , SAS = 0 AVDD = 3.3V, I_{AVDD} ≤5mA, OVDD = 3.3V, I_{OVDD} ≤70mA, OVSS = -4.7V, I_{OVSS} ≤70mA	–20	-	+20	mV
Soft Start Time	t_{SS2}		-	2	-	ms
Under Voltage Protect Voltage	V_{UVP_AVDD}		-	0.8x V_{AVDD}	-	V
Duration to UVP Trigger Time	t_{UVP_AVDD}		-	1	-	ms

Electrical Characteristics

(V_{IN} = 2.9V to 4.6V, T_A = 25℃ , unless otherwise specified. V_{IN} = 3.7V, SAS = 0, SRL=pulse7, V_{AVDD} = 3.3V, V_{OVDD} = 3.3V, V_{OVSS} = −3.3V, C_{INA} = C_{fly1} = C_{fly2} = C_{VIN} = C_{AVDD} = C_{OVDD} = C_{BST} = C_{VON} = 10μF, C_{OVSS} = 20μF)

OVDD

PARAMETER	SYMBOL	TEST CONDITION	MIN	TYP	MAX	UNIT
Output Voltage Range	V_{OVDD}	SAS = 0	2.8	3.3	4.6	V
Output Voltage Range		SAS = high		4.6 (fixed)		V
Output Voltage Tolerance		V_{OVDD}(default) I_{OVDD} = 10mA	−0.5	-	+0.5	%
Load Current Capacity	I_{OVDD}	SAS = 0 AVDD = 3.3V, OVDD = 3.3V, OVSS = –3.3V	50	-	-	mA
		SAS = High AVDD = 2.8V, OVDD =4.6V, OVSS = –2.2V				
		SAS = 0 V_{OVDD} = 4.5V, V_{OVSS} = –4.5V	70	-	-	mA
Line Regulation V_{IN} = 2.9V to 4.6V, SAS = 0		OVDD = 3.3V, I_{OVDD} = 50mA	-	5	10	mV
		OVDD = 4.5V, I_{OVDD} = 80mA				
Load Regulation		SAS = 0 OVDD = 3.3V I_{OVDD} = 0mA to 50mA	-	5	10	mV
		SAS = 0 OVDD = 4.5V I_{OVDD} = 0mA to 80mA				
Output Transient Ripple		V_{IN} = 3.7V SAS = 0 , AVDD = 3.3V, OVDD = 3.3V, OVSS = −3.3V I_{AVDD} & I_{OVDD} & I_{OVSS} ≤ 5mA	–10	-	+10	mV
		VIN = 3.7V SAS = high , AVDD = 2.8V, OVDD = 4.6V, OVSS = −2.2V I_{AVDD} & I_{OVDD} & I_{OVSS} ≤ 5mA				
Output Voltage Ripple		V_{IN} = 3.7V, SAS = 0 AVDD = 3.3V, I_{AVDD} ≤5mA OVDD = 3.3V, I_{OVDD} ≤ 30mA OVSS = -3.3V, I_{OVSS} ≤ 30mA	−10	-	+10	mV
		V_{IN} = 3.7V, SAS = high AVDD = 2.8V, I_{AVDD} ≤ 5mA OVDD = 4.6V, I_{OVDD} ≤ 30mA OVSS = -2.2V, I_{OVSS} ≤ 30mA				

OVDD

PARAMETER	SYMBOL	TEST CONDITION	MIN	TYP	MAX	UNIT
Output Voltage Ripple		V_{IN} = 3.7V , SAS = 0 AVDD = 3.3V, I_{AVDD} ≤5mA, OVDD = 3.3V, I_{OVDD} ≤70mA, OVSS = -4.7V, I_{OVSS} ≤70mA	−10	-	+10	mV
Soft Start Time	t_{SS3}		-	2	-	ms
Under Voltage Protect Voltage	V_{UVP_OVDD}		-	0.8x V_{OVDD}	-	V
Duration to UVP Trigger Time	t_{UVP_OVDD}		-	1	-	ms

■ Electrical Characteristics

(V_{IN} = 2.9V to 4.6V, T_A =25℃ , unless otherwise specified. V_{IN} = 3.7V, SAS = 0, SRL=pulse7, V_{AVDD} = 3.3V, V_{OVDD} = 3.3V, V_{OVSS} = −3.3V, C_{INA} = C_{fly1} = C_{fly2} = C_{VIN} = C_{AVDD} = C_{OVDD} = C_{BST} = C_{VON} = 10μF, C_{OVSS} = 20μF)

Negative Charge Pump Controller for VON

PARAMETER	SYMBOL	TEST CONDITION	MIN	TYP	MAX	UNIT
VON Output Voltage	V_{VON}	SAS = 0	–5.5	-	-	V
		SAS = high	–2.6	-	-	V
VON Maximum Load Current	I_{VON_MAX}	SAS = 0 V_{OVDD} = 3.3V, V_{OVSS} = –3.3V	62.5	-	-	mA
		SAS = 0 V_{OVDD} = 4.5V, V_{OVSS} = –4.5V	100	-	-	mA
Negative Charge Pump Clock Frequency	f_{OSC_VON}			750	-	KHz

OVSS

PARAMETER	SYMBOL	TEST CONDITION	MIN	TYP	MAX	UNIT
Output Voltage	V_{OVSS}	SAS = 0	−5.0	−2.2	−0.6	V
Output Voltage		SAS = high	−2.4	−2.2	−0.6	V
Output Voltage Tolerance		V_{OVSS}(default) I_{OVSS} = 10mA	–1	-	+1	%
Load Current Capacity	I_{OVSS}	SAS = 0 AVDD = 3.3V, OVDD = 3.3V, OVSS = – 3.3V SAS = high AVDD = 2.8V, OVDD =4.6V, OVSS = – 2.2V	50	-	-	mA
		SAS = 0 V_{OVDD} = 4.5V, V_{OVSS} = –4.5V	70	-	-	mA

Output Transient Ripple		V_{IN} = 3.7V SAS = 0 , AVDD=3.3V, OVDD=3.3V, OVSS = -3.3V I_{AVDD} & I_{OVDD} & I_{OVSS} ≤ 5mA	–10	-	+10	mV
		V_{IN} = 3.7V SAS = high , AVDD=2.8V, OVDD=4.6V, OVSS = -2.2V I_{AVDD} & I_{OVDD} & I_{OVSS} ≤ 5mA				
Output Voltage Ripple		V_{IN} = 3.7V, SAS = 0 AVDD = 3.3V, I_{AVDD} ≤5mA OVDD = 3.3V, I_{OVDD} ≤30mA OVSS = -3.3V, I_{OVSS} ≤30mA	–20	-	+20	mV
		V_{IN} = 3.7V, SAS = high AVDD = 2.8V, I_{AVDD} ≤ 5mA OVDD = 4.6V, I_{OVDD} ≤ 30mA OVSS = -2.2V, I_{OVSS} ≤ 30mA				
		V_{IN} = 3.7V , SAS = 0 AVDD = 3.3V, I_{AVDD} ≤5mA, OVDD = 3.3V, I_{OVDD} ≤70mA, OVSS = -4.7V, I_{OVSS} ≤70mA				
Soft Start Time	t_{SS5}		-	2	-	ms
Under Voltage Protect Voltage	V_{UVP_OVSS}		-	0.4x V_{OVSS}	-	V
Duration to UVP Trigger Time	t_{UVP_OVSS}		-	1	-	ms

Input Control Block

OCP21351 detects the input voltage, from the battery, to determine whether the boost regular should be activated. In addition, OCP21351 detects SAS, EN and SRL pins status to decide whether it should be in operation or shutdown.

AVDD

OCP21351 outputs 2.6V to 3.6V (one-wire program) AVDD voltage through a built in LDO, with 2ms soft start time and minimum 10mA output current. Figure 1 AVDD block diagram for your reference

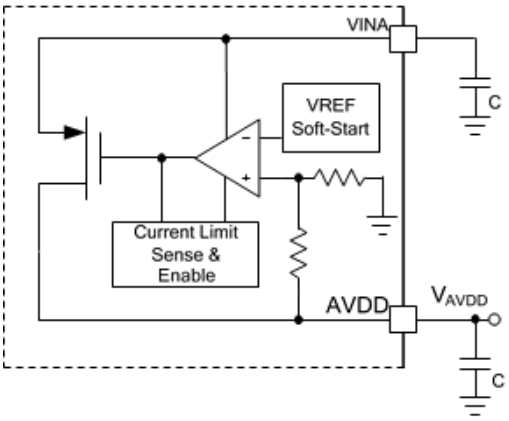


Figure 1. AVDD Block Diagram

OVDD

OCP21351 outputs OVDD voltage through a built in LDO, with minimum 50mA output current.

In symmetry mode (SAS = Low), OVDD output voltage can be set between 2.8V to 4.6V, with 0.1V resolution, through Serial Data programing scheme. Block diagram is shown in Figure 2 for your reference.

In asymmetry (SAS = high), OVDD should be fixed at initial value, 4.6V. Any OVDD setting change is not allowed

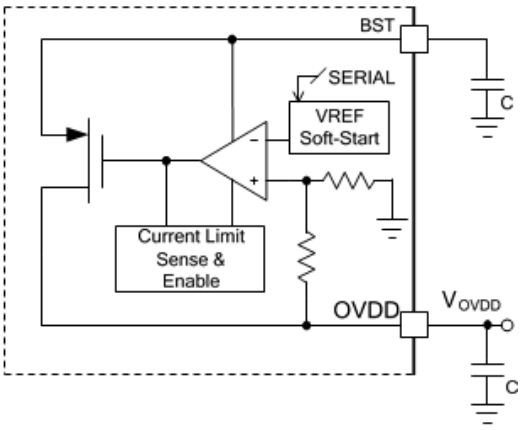


Figure 2. OVDD Block Diagram

VON

In order to generate the negative voltage output for OVSS, OCP21351 integrates a negative charge pump controller, along with two external flying capacitors, to invert either battery VIN or boost output VBOOST as the input source of LDO for OVSS. An external 10µF stabilizing capacitor is strongly recommended on this VON pin. Block diagram is shown in Figure 3. for your reference

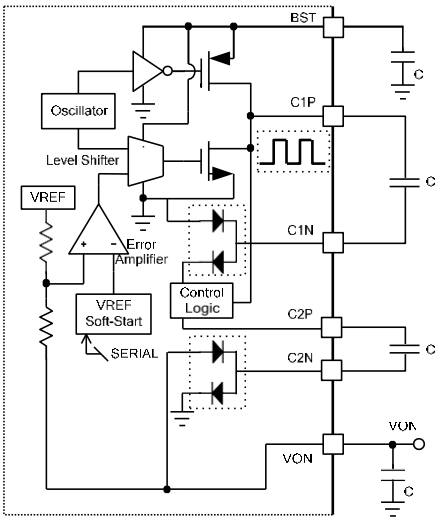


Figure 3. VON Block Diagram

OVSS

OCP21351 outputs OVSS voltage through a built in LDO, with minimum 50mA output current. OVSS output voltage is programmable from −0.6V to −5.0V with 0.1V resolution through Serial Data programing scheme. Block diagram is shown in Figure 4 for your reference.

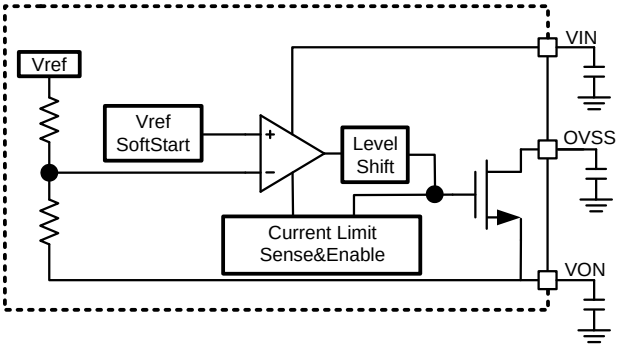


Figure 4.OVSS Block Diagram

SAS, Symmetry or Asymmetry power Select pin

VAVDD can be changed to 2.8V or 3.3V by SAS pin. When SAS is pulled low, AVDD is set to 3.3V. When SAS is pulled high, AVDD is set to 2.8V.

If SAS is set to Low, OCP21351 is defined to operate at Symmetry mode. On other hand, If SAS is set to High, OCP21351 is defined to operate at Asymmetry mode.

Please make sure this pin setting should be fixed before OCP21351 starts operation, before EN goes to High.

UVLO (Under Voltage Lockout)

To prevent the battery from being completely drained, OCP21351 will automatically shut down when the input UVLO falling threshold of 2.20V is detected. OCP21351 will restart to work automatically as soon as the input voltage exceeds back to the UVLO rising threshold of 2.60V.

Under Voltage Protection (UVP)

To prevent any abnormal condition to damage the panel system, OCP21351 employees UVP protection in all 3 LDOs. As soon as the output drops below 80% of AVDD, or 80% of OVDD, or 40% of OVSS, OCP21351 will shut down all outputs to GND. After UVP, OCP21351 can be back to function by toggling EN pin.

Thermal Shutdown

To prevent the overheat operation, OCP21351 will shut down the outputs when approximately 150°C of junction temperature is detected by OCP21351. After the thermal shutdown occurs, OCP21351 will restart the output functions as soon as the junction temperature cools down approximately 20°C lower, ~130°C.

Power Sequence

1. OCP21351 starts operation when VIN exceeds 2.6V; and stops as soon as VIN drops below 2.2V.
2. The SAS pin defines the default value of AVDD and OVDD, so it should be pull high or low level before EN pin is driven to High level.
3. As soon as EN is driven to High level, OCP21351 will drive AVDD to 3.3V or 2.8V, followed by SAS's setting, along with soft start behavior. Meanwhile, SRL should be remained at Low. And the OVDD and OVSS outputs keep at GND level.
4. OVDD will be driven to default setting value of voltage after SRL input is detected as High level by OCP21351. As well, OVSS will be also driven to default setting value of voltage after OVDD. Both outputs include soft start function.
5. Keep SRL as High for 6 ms at least, then Serial Date programming scheme can be proceeded to adjust the OVDD and OVSS to their target value
6. Whenever EN is pulled to Low level, all the outputs will be pulled back to GND

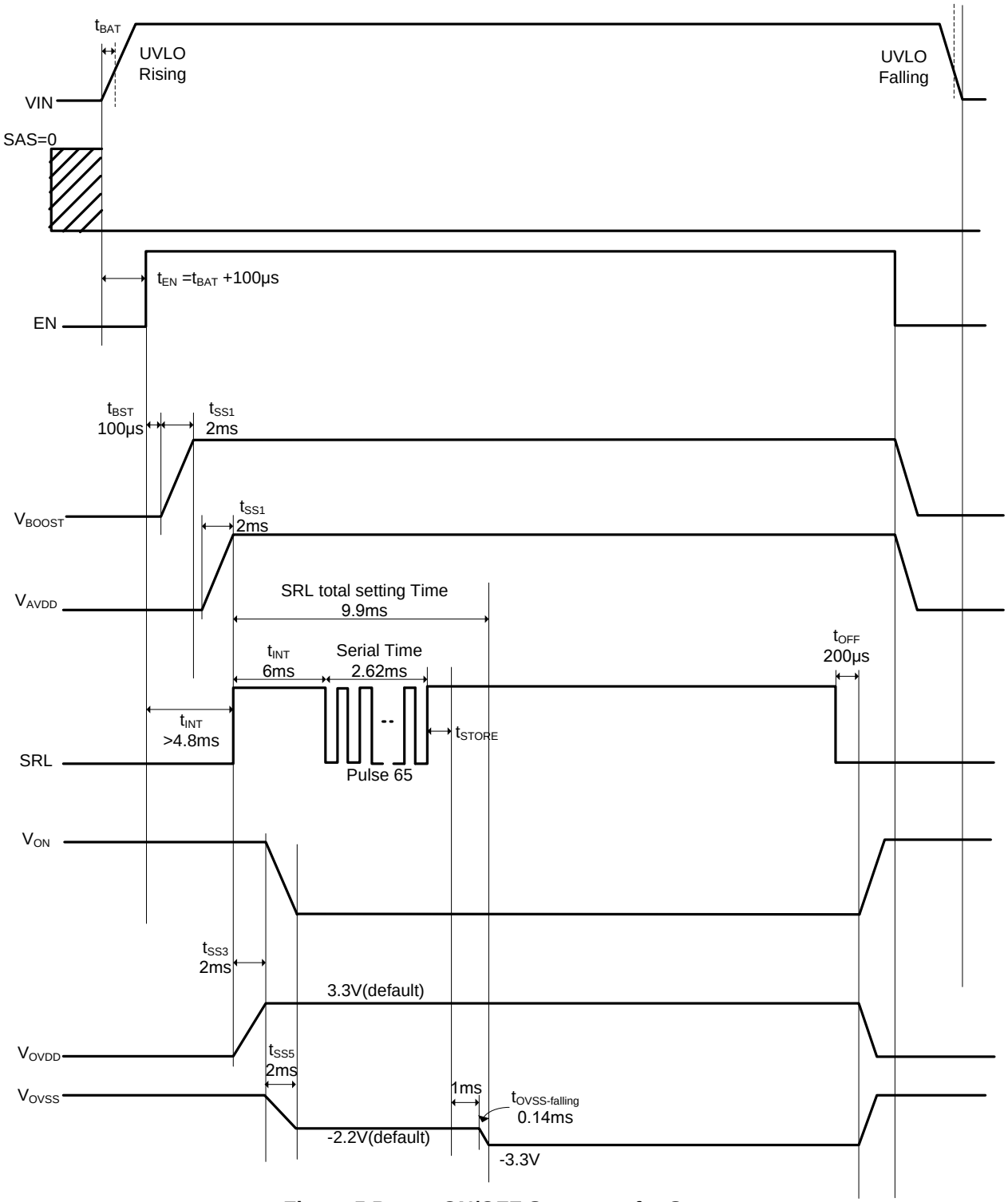


Figure 5.Power ON/OFF Sequence for Symmetry

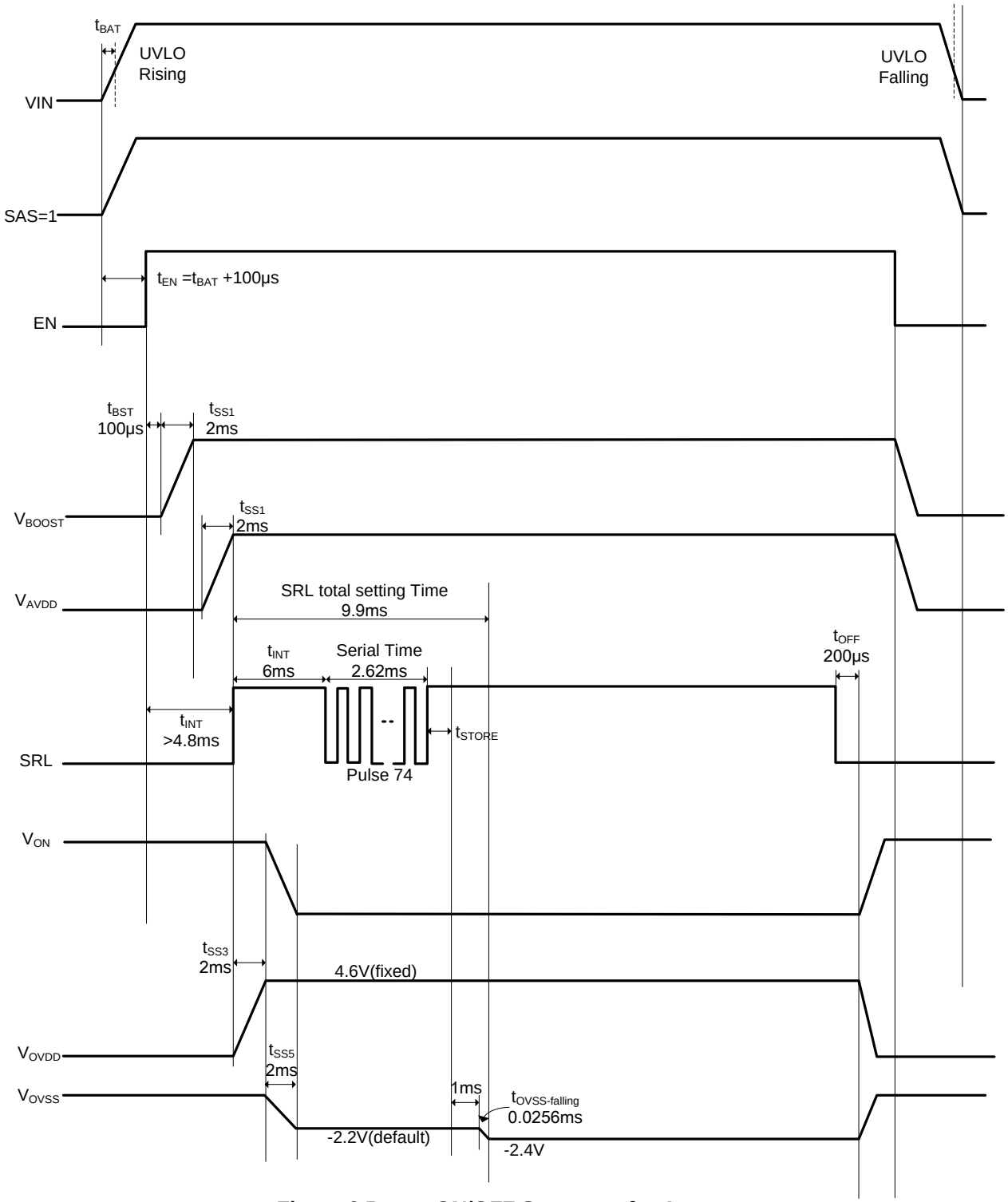


Figure 6.Power ON/OFF Sequence for Asymmetry

Battery Power Saving Options

- **Shutdown by HW control, EN & SRL**

  OCP21351 will enter shutdown mode and only consume 1μA of input current when EN and SRL pins are driven to Low. Therefore, system controller can save the battery power consumption by force OCP21351 into shutdown mode when display is allowed to be off.

- **Idle Mode by SRL Setting**

  During Light Load operation on panel system, OCP21351 can be programmed into Idle mode to consume lower input current from battery by turning off some unnecessary protection and lowing switch frequency, while remaining AVDD and OVDD at 3.3V, as well as OVSS at –3.3V.

Serial Data Interface

OCP21351 SRL is a serial data input pin with a pull low 700kΩ internal resistor. The low pulses count on SRL is the protocol to set OVDD/OVSS target output voltage, as well as the mode selection

Figure 7 shows the SRL timing and how the OVDD/OVSS responds after the SRL protocol settings. Table 1. Lists the timing parameter requirement of SRL for your reference.

Table 2. Lists the SRL pulse count mapping table for your reference.

Please be noted that both OVDD and OVSS takes 0.1V and 128μs per step approaching to target.

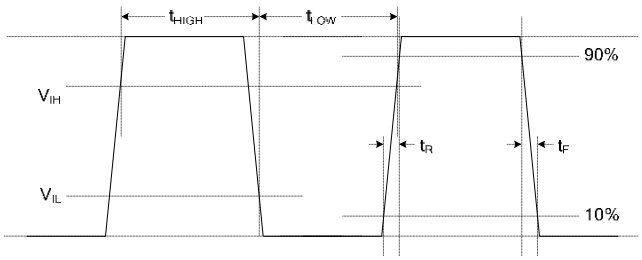


Figure 7. SRL Waveform Chart

Table 1. SRL Timing Parameters

SYMBOL	DESCRIPTION	MIN	TYP	MAX	UNIT
t_R	SRL Rise Time	-	-	200	ns
t_F	SRL Fall Time	-	-	200	ns
V_{IH}	SRL Input High Voltage	1.2	-	V_{IN}	V
V_{IL}	SRL Input Low Voltage	0	-	0. 4	V
t_{HIGH}	SRL High Level Time Period	8	20	80	μs
t_{LOW}	SRL Low Level Time Period	8	20	80	μs
t_{INT}	SRL Initialization Time	-	6	-	ms
t_{STORE}	End of Pulse Count	140	-	-	μs
t_{OFF}	SRL OFF	200	-	-	μs
t_{NEXT}	Next Data Waiting Time	8		20	ms
t_{STEP}	OVDD/OVSS Step Time	-	128	-	μs

Table 2. SAS = high for Asymmetry

PULSE	FUNCTION DESCRIPTION						
1	reserved						
2	reserved						
3	reserved						
4	reserved						
5	Fast discharge function on						
6	reserved						
7	reserved						
8	reserved						
9	reserved						
74 ~ 92	OVSS setting, –2.4V ~ –0.6V						
116	OVDD turn on						
117	VON/OVSS turn on						
118	OVDD turn off						
119	VON/OVSS turn off						
120	OVSS discharge to GND, VOVSS = 0V						
121	reserved						
128~138	AVDD setting, 2.6V ~ 3.6V						
PULSE	**OVDD (V)**	**PULSE**	**OVSS (V)**	**PULSE**	**OVSS (V)**	**PULSE**	**AVDD(V)**
10	reserved	48	reserved	71	reserved	128	2.6
11		49		72		129	2.7
12		50		73		130	2.8
13		51		74	−2.4	131	2.9
14		52		75	−2.3	132	3.0
15		53		76	−2.2	133	3.1
16		54		77	−2.1	134	3.2
17		55		78	−2.0	135	3.3
18		56		79	−1.9	136	3.4
19		57		80	−1.8	137	3.5
20		58		81	−1.7	138	3.6
21		59		82	−1.6	-	-
22		60		83	−1.5	-	-
23		61		84	−1.4	-	-
24		62		85	−1.3	-	-
25		63		86	−1.2	-	-
26		64		87	−1.1	-	-
27		65		88	−1.0	-	-
28		66		89	−0.9	-	-
-	-	67		90	−0.8	-	-
-	-	68		91	−0.7	-	-
-	-	69		92	−0.6	-	-
-	-	70		-	-	-	-

Table 3. SAS = low for Symmetry

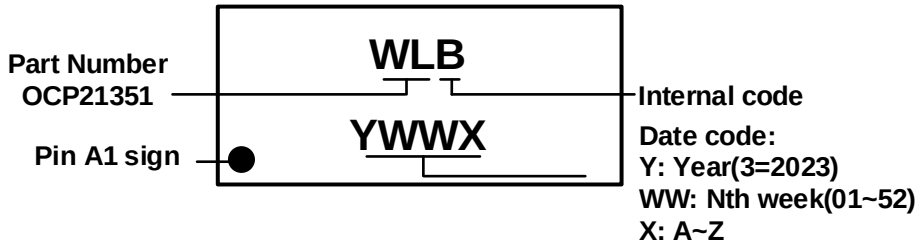
PULSE	FUNCTION DESCRIPTION						
1	reserved						
2	reserved						
3	reserved						
4	reserved						
5	Fast discharge function on						
6	reserved						
7	OVDD = 3.3V, OVSS = –3.3V						
8	OVDD = 4.0V, OVSS = –4.0V						
9	OVDD = 4.5V, OVSS = –4.5V						
74 ~ 92	OVDD setting, 2.8V ~ 4.6V						
116	OVSS setting, –5V ~ –0.6V						
117	OVDD turn on						
118	VON/OVSS turn on						
119	OVDD turn off						
120	VON/OVSS turn off						
121	OVSS discharge to GND, VOVSS = 0V						
128~138	AVDD setting, 2.6V ~ 3.6V						
PULSE	**OVDD (V)**	**PULSE**	**OVSS (V)**	**PULSE**	**OVSS (V)**	**PULSE**	**AVDD(V)**
10	2.8	48	−5.0	71	−2.7	128	2.6
11	2.9	49	−4.9	72	−2.6	129	2.7
12	3.0	50	−4.8	73	−2.5	130	2.8
13	3.1	51	−4.7	74	−2.4	131	2.9
14	3.2	52	−4.6	75	−2.3	132	3.0
15	3.3	53	−4.5	76	−2.2	133	3.1
16	3.4	54	−4.4	77	−2.1	134	3.2
17	3.5	55	−4.3	78	−2.0	135	3.3
18	3.6	56	−4.2	79	−1.9	136	3.4
19	3.7	57	−4.1	80	−1.8	137	3.5
20	3.8	58	−4.0	81	−1.7	138	3.6
21	3.9	59	−3.9	82	−1.6	-	-
22	4.0	60	−3.8	83	−1.5	-	-
23	4.1	61	−3.7	84	−1.4	-	-
24	4.2	62	−3.6	85	−1.3	-	-
25	4.3	63	−3.5	86	−1.2	-	-
26	4.4	64	−3.4	87	−1.1	-	-
27	4.5	65	−3.3	88	−1.0	-	-
28	4.6	66	−3.2	89	−0.9	-	-
-	-	67	−3.1	90	−0.8	-	-
-	-	68	−3.0	91	−0.7	-	-
-	-	69	−2.9	92	−0.6	-	-
-	-	70	−2.8	-	-	-	-

■ Ordering Information

Part Number	Driver Capability	Package Type	Package Qty	Temperature	Eco Plan	Lead/Ball Finish
OCP21351WPAD	100mA	21-Ball WLCSP	3000pcs	-40～85℃	Green	Sn/Ag/Cu

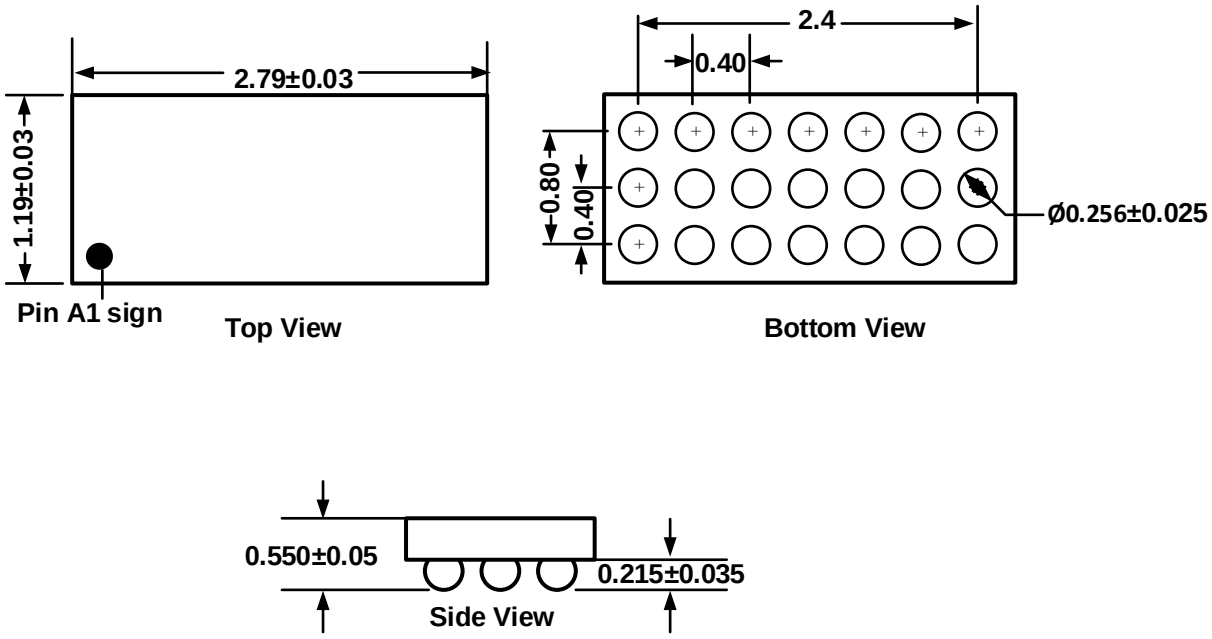
■ Marking Information

21-Ball WLCSP



■ Package Information

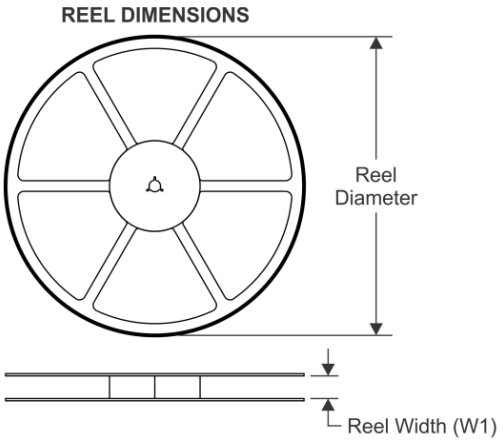
21-Ball WLCSP

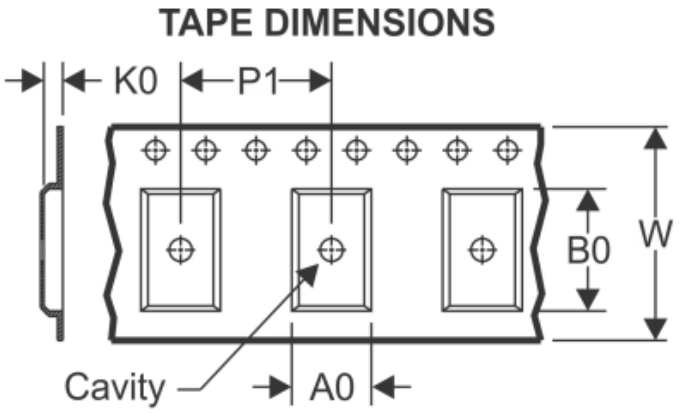


NOTE: All dimensions are in mm.

■ Packing Information

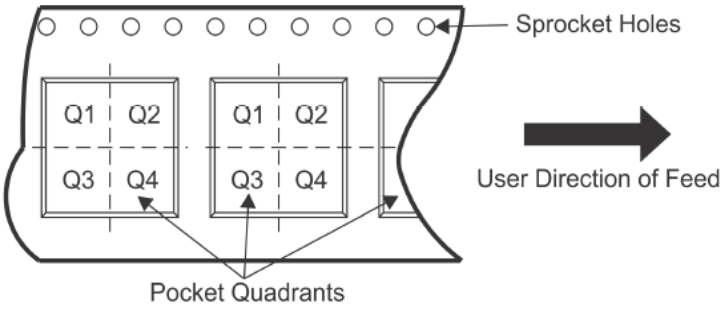
21-Ball WLCSP





A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



Package tape	Package Drawing	MSL	SPQ	Reel Diameter (mm)	Reel Width W1(mm)
21-Ball WLCSP	WLB	Level-1-260℃	3000	330	12.5
A0 (mm)	**B0 (mm)**	**K0 (mm)**	**P1 (mm)**	**W (mm)**	**PIN A1 Quadrant**
1.4	3.0	0.72	4.0	12	Q1

IMPORTANT NOTICE

Orient-Chip Semiconductor Co., Ltd (OCS) and its subsidiaries reserve the right to make corrections, modifications, enhancements, improvements, and other changes to its products. Customers should obtain the latest relevant information before placing orders and should verify that such information is current and complete. This separate provision won't be provided.